EXPERIMENT 12

Port Scanning Tools

AIM : To perform port scanning in Kali Linux using Nmap and identify open, closed, and filtered ports on a target system.

DESCRIPTIONS

1.nmap –version

This command displays the installed Nmap version in Kali Linux.

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcre2-10.46 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

2. nmap 127.0.0.1

This command scans common ports on the local computer and shows their status.

```
┌──(kali㉿kali)-[~]
└─$ nmap 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-08 23:12 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000010s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

3. nmap -p 22,80 127.0.0.1

Scans only ports 22 and 80 on the local computer.

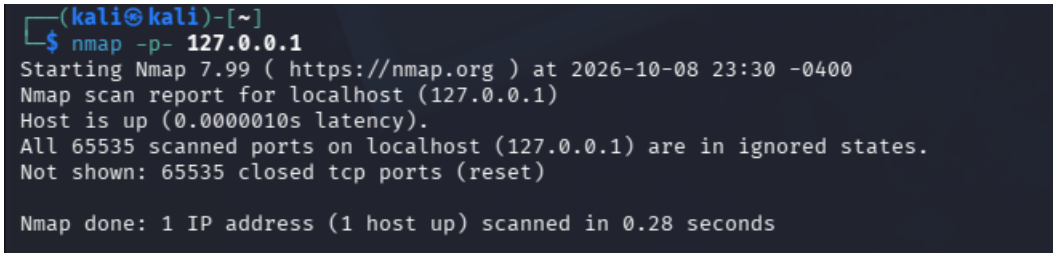
```
┌──(kali㉿kali)-[~]
└─$ nmap -p 22,80 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-08 23:26 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000066s latency).

PORT   STATE  SERVICE
22/tcp closed ssh
80/tcp closed http

Nmap done: 1 IP address (1 host up) scanned in 0.04 seconds
```

4.nmap -p- 127.0.0.1

Scans all TCP ports from 1 to 65535.



RESULT : Port scanning was demonstrated using Nmap in Kali Linux. The scan reports the target's reachable ports and their states.